

Graded Support v2 Audit Note

March 21, 2026

Status

This note records the current status of the graded-support sandbox after the v2 rewrite of graded expression bookkeeping.

The main result is mixed:

The graded-support ontology still looks better than the binary ontology, and the v2 rewrite improved bookkeeping discipline, but the selector hierarchy is still not right.

More specifically, the v2 sandbox remains expression-led rather than constraint-led in local move selection, and late **edge_up** moves remain too easy.

1 Purpose

The purpose of the v2 rewrite was to replace the original graded-support bonus layer with a more disciplined, witness-based graded expression scheme.

The intended improvement was:

1. keep graded subsystem commitment

$$\sigma_i \in [0, 1],$$

and graded interface commitment

$$w_{ij} \in [0, 1],$$

2. but remove the earlier flat graded bonuses,
3. and make move selection track faithful differentiated local expression more directly.

So the key question for v2 was not whether graded support still produced partial carriers. The key question was whether the local selector had become more faithful to the four lawful pressures.

2 What did not change

At the coarse regime level, v2 did *not* change the main outcome.

The run still produced:

- 6 accepted **raise_support** moves,
- 20 accepted **edge_up** moves,

- 4 accepted `edge_down` moves,
- 0 accepted `lower_support` moves.

The final support commitments remained:

$$\sigma = (1, 1, 0.35, 0.35, 0.35, 0.35, 0.35, 0.35),$$

so the final support picture still consists of two full carriers and six persistent partial carriers.

The system still reaches full site occupancy and then settles into a late regime of fixed support mass with interface retuning rather than support demotion.

So the coarse pattern remains:

seed partial carriers $\rightarrow$ reach full site occupancy $\rightarrow$ late interface retuning.

3 What did change

The bookkeeping form improved.

In particular:

1. the v2 rewrite no longer relies on the earlier flat graded-bonus layer in the same direct way,
2. accepted `edge_down` moves can still win strongly in the middle regime,
3. late `edge_up` wins are now much smaller than in the earlier graded-support pass.

So the v2 rewrite improved the *form* of the bookkeeping even though it did not yet change the *regime*.

That is real progress, but it is not yet enough.

4 Constraint-probe reading

The v2 constraint probe preserves the same broad interpretation as the earlier graded-support pass.

The current best broad reading is:

1. finite bandwidth explains why support mass can remain broadly distributed across many partial carriers,
2. no-refolding explains why the resulting late-time geometry becomes sticky,
3. no-forgetting explains why the partial carriers persist once they have formed,
4. no-signalling explains why the late dynamics become local interface retuning rather than more support raising.

The probe numbers remain strong:

- 6 partial nodes,
- diffuse partial fraction ≈ 0.512 ,

- first full occupancy at step 100,
- post-full partial persistence fraction = 1.0,
- post-full full-node stability fraction = 1.0,
- late edge Jaccard mean = 1.0,
- late sigma L^1 drift = 0.0,
- late move flow 100% `edge_up`.

So the broad regime is still compatible with the four lawful pressures.

5 Move-audit result

The move audit is the decisive diagnostic.

Its central result is:

Selection is still expression-led, constraint-filtered, not constraint-led.

That is the most important current status statement.

Early regime

In the early support-raising regime, accepted `raise_support` moves still have large positive

$$dE_{\text{expr}},$$

while the constraint terms are much smaller in magnitude.

Among the lawful penalties, no-signalling is often the largest active early-time filter, but it is still only a filter on top of a strongly positive expression term.

Middle regime

The middle regime still allows strong `edge_down` winners. That is good. It means the system is not incapable of accepting lawful demotion-like or relief-releasing interface changes.

But these do not yet lead to actual accepted `lower_support` moves.

Late regime

The late regime remains the key problem.

From the late accepted rows, the structure is:

1. `edge_up` keeps winning,
2. $dE_{\text{expr,raw}}$ is slightly negative,
3. but total dE_{expr} remains slightly positive,
4. constraint penalties are tiny,
5. so the move survives because the witness-based expression correction is still net positive.

Thus, even after the v2 rewrite, the model still grants enough expression-side credit to late interface accretion that it outruns all demotion paths.

6 What this means

The current graded-support program should therefore be understood as follows.

1. The ontology shift still looks right

The move from binary support to graded support remains promising.

It is still better than:

- pure binary support occupancy,
- repeated same-child birth/retire churn as the only way to express intermediate demand,
- or proto-generalization collapsing into a one-parent star.

So the graded-support ontology should not be abandoned.

2. The selector hierarchy is still wrong

The real unresolved problem is not the existence of partial carriers.

The unresolved problem is:

the selector still rewards interface accretion too easily once a partial-support network has already formed.

This is why:

- finite bandwidth is still not functioning as a genuine local selector,
- no-refolding mainly appears as late-time stickiness rather than discriminating early support choices,
- no-forgetting appears mainly as persistence of already-formed carriers,
- and late `edge_up` still survives because expression remains slightly positive even when raw local gain is already slightly negative.

3. The current failure is now sharper

This is progress, because the remaining failure is now more precise than before.

It is no longer:

The binary ontology is too coarse.

It is now:

The graded ontology is probably right, but graded expression is still too generous to late interface accretion.

That is a much sharper scientific target.

7 Constraint-faithful interpretation

Because only four lawful pressures are available, the next step cannot be to add a fifth pressure or invent a new penalty by hand.

The next step must instead be:

make graded expression more faithful to the four existing pressures.

In particular, late **edge_up** should not remain acceptable merely because persistence-like terms keep its corrected expression slightly positive.

If a late **edge_up** move has:

$$dE_{\text{expr,raw}} < 0$$

and no genuine differentiated-role gain, then persistence alone should not rescue it.

Persistence belongs under no-forgetting as a conservation-style reality condition. It should not act like an automatic local reward for adding yet another interface.

8 Immediate consequence

The immediate consequence is:

graded expression for **edge_up** must be tightened so that it earns expression credit only from real differentiated-role gain or real parent/interface relief improvement, not from tolerated persistence alone.

That is the next coding target.

9 Next implementation question

The next implementation question should therefore be stated explicitly as:

Can graded expression be reformulated so that late **edge_up** only remains positive when it produces genuine local differentiated-role gain or genuine interface-relief gain, rather than merely preserving an already-existing partial-support structure?

That is the right next derivation problem.

One-sentence summary

The v2 rewrite improved the graded-support bookkeeping, but it did not yet change the regime: the sandbox still settles into persistent diffuse partial carriers with late interface retuning, and the remaining failure is now sharp—graded expression is still too generous to late **edge_up** moves even after raw local gain has already turned slightly negative.